Hachion

Module 1: Introduction to Azure DevOps

- Overview of DevOps and its importance
- Introduction to Azure DevOps and its key components
- Understanding Agile and DevOps methodologies
- Setting up an Azure DevOps account

Module 2: Azure Repos (Version Control)

- Introduction to version control and Git
- Setting up and managing repositories in Azure Repos
- Branching strategies and pull requests
- Code review and collaboration in Azure Repos

Module 3: Azure Pipelines (CI/CD)

- Introduction to Continuous Integration (CI) and Continuous Deployment (CD)
- Creating and managing build pipelines
- YAML vs. Classic pipelines
- Running automated tests in Azure Pipelines
- Deploying applications to Azure services

Module 4: Azure Test Plans (Testing & Quality Assurance)

- Overview of test management in Azure DevOps
- Writing and executing test cases
- Manual vs. automated testing
- Performance and load testing

Module 5: Azure Boards (Agile Project Management)

- Introduction to work items, epics, features, and user stories
- Creating and managing sprints
- Kanban boards and backlog management
- Tracking progress with dashboards and reports

Module 6: Azure Artifacts (Package Management)

- Introduction to package management in DevOps
- Creating and managing Azure Artifacts
- Publishing and consuming packages
- Integrating with other DevOps tools

Module 7: Infrastructure as Code (IaC) with Azure DevOps

- Introduction to Infrastructure as Code
- Using ARM templates and Terraform in Azure DevOps
- Automating infrastructure deployment
- Configuration management with Ansible and Azure DevOps

Module 8: Security & Compliance in Azure DevOps

- Implementing security best practices
- Managing permissions and access control
- Secrets management with Azure Key Vault
- Compliance and auditing in Azure DevOps

Module 9: Monitoring & Logging in Azure DevOps

- Introduction to monitoring tools in Azure
- Setting up alerts and dashboards
- Log analytics and troubleshooting
- Performance tuning in DevOps workflows

Module 10: Advanced DevOps Practices & Real-World Use Cases

- Blue-Green and Canary deployments
- Microservices and containerization with Kubernetes & Docker
- DevOps automation with GitHub Actions and Azure DevOps
- Case studies and industry best practices